YEAR 8 · SCIENCE DETECTIVES 科学侦探 · WEEK 7 OF 10 · 40 MINUTES

# Design your own test 设计你的检测

**A good forensic test has a clear question, tests the scene sample exactly like the references, changes one thing and is planned to be safe.**

- I can write a testable question and name what we change, observe or measure, and keep the same.
- I can plan a safe method with reference samples, repeats, a risk table and a results table.

**testable question** 可检验的问题 · **independent variable** 自变量 · **dependent variable** 因变量 · **control variable** 控制变量 · **fair test** 公平测试 · **reference sample** 参照样本 · **method** 实验步骤 · **prediction** 预测 · **hazard** 危险源 · **risk** 风险

## THE LESSON

| | | |
|---|---|---|
| **Do Now** | 4 | Three recap questions on whiteboards or in the logbook; Today slide with goals and key words |
| **Hook** | 3 | Show sealed Evidence 6 and 7 bags; hand out option cards; Reporters read the question aloud |
| **Learn** | 7 | Six planning steps modelled with the Week 4 powder test; one worked risk-table row (vinegar) |
| **Do** | 18 | Groups plan on the plan sheet; checkpoints at minutes 6 and 12; results table and kit request |
| **Think and Reflect** | 8 | Pick the fairer soil method; plan swap (skip if short); exit ticket; collect sheets at the door |

**Hands-on:** No practical: groups plan their Week 8 test (Soil, Paper, Ink 2 or Powder 2) and only look at sealed bags.

**Collect or assess:** Plan sheet (rubric strands 1 and 2) signed off at checkpoints 1 and 2; exit ticket; kit requests; evidence log row 7.

## SAFETY

- No practical work: evidence bags stay sealed and go back to the teacher.
- Do not open, smell or taste any sample.
- Teacher checks every group's risk table before any Week 8 practical.

## WATCH OUT

- Groups change two things, list kit instead of a method, or skip repeats: insist on the 'We change' frame.
- Evidence 6 must come from flowerbed A; Evidence 7 from the lab's own roll, arrow marked before crumpling.

**Language:** Frames: 'We change / observe / keep ___'. Hazard is the thing, risk the chance. Band 1: language buddy, Core b) first.

**Next week:** Week 8 lab day: commented plan sheets back in the Do Now; keep option cards and bags; trays from kit requests.

## PREP FOR ONE CLASS

**before the lesson about 50 min (10 could be done by someone else); after about 43 min (some times estimated)**

### 2 OR MORE WEEKS BEFORE

- [ ] Book printing: plan sheets x24 (48 sheets), kit card, Week 8 kit lists 5 min · reprographics

### THE WEEK BEFORE

- [ ] Allocate options, print 10 option cards, make Evidence 6 and 7 bags 30 min · teacher
- [ ] Evidence 3 bags: unopened Week 4 bag, or about 3 tsp baking soda each 5 min · technician
- [ ] Note ink strips N: Week 3 spares or new with pen B, in clear sleeves 5 min · teacher
- [ ] Laminate the 10 option cards (optional) optional 10 min · technician
- [ ] Collect reference soils A to D and papers P1 to P4 (optional) optional 45 min · technician

### ON THE DAY, BEFORE THE LESSON

- [ ] Set out group kits, class box items; Evidence 6 and 7 on teacher desk 5 min · teacher

### AFTER THE LESSON

- [ ] Clean-up: bags and N strips to teacher, keep option cards, wipe boards 3 min · students
- [ ] Read plan sheets: check risk tables, one comment each, read exit tickets 30 min · teacher
- [ ] Turn the six kit requests into Week 8 tray lists 10 min · teacher

## PRINT

- Plan sheet x24 (4 A4 pages: 2 sheets each, double-sided)
- Option cards x10 (1 sheet each, single-sided: 6 group, 4 spare)
- Teacher copies: kit card (A5), Week 8 kit lists by option (4 A5)

## GET OR MAKE (NEW THIS LESSON)

- Sealed evidence bags, up to 2 each: Evidence 6, 7, 3
- Note ink strips N x2 (Week 3 spares first)
- Small zip bags x6
- Sticky labels x6
- Clear plastic sleeves x2
- Lab paper towel sheets x2 (from the lab's own roll)
- Baking soda 6 tsp, if no sealed Week 4 bag (optional)
- Coffee-filter strips 2 cm x 10 cm x2 (optional)
- Mini whiteboards x24 (optional)
- Dry-wipe pens x24 (optional)
- Laminating pouches A4 x10 (optional)
- Reference soils A to D, papers P1 to P4 (optional)

**Have ready from earlier weeks:** Class box, Case files x24 (Week 4 results grid), Scientist's Logbooks x24, Role card sets x6 (with buddy card), Sentence frame card sets x6, My Science Dictionary x24, Pencils x24, Pen B, for new N strips (optional)